Abishek Soti

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PROFILE

Machine Learning researcher and engineer focused on compact, high-performing models for audio, computer vision, neuromorphic sensing, and edge deployment. Experience spans feature engineering, classical ML, CNNs, Transformers, biologically inspired signal representations, model optimisation, and deployment using Python, PyTorch, scikit-learn, Raspberry Pi, and Hugging Face.

TECHNICAL SKILLS

Machine Learning: PyTorch, scikit-learn, CNNs, Vision Transformers, EfficientNetB2, RBF-SVM, MLPs, feature engineering, model evaluation, classification analysis
Audio & Signal Processing: STFT, Mel spectrograms, MFCC, CQT, statistical pooling, CARFAC, BM, IHC, LI, spike-derived features, DSP
Programming & Data: Python, NumPy, pandas, MATLAB, SQL, SQLite, Jupyter, Matplotlib, Git, GitHub, Linux, Bash, basic C/C++
Deployment & Systems: Raspberry Pi, Hugging Face, Docker, AWS EC2/ECR/Elastic Beanstalk, Azure Containers, Arduino, embedded-system workflows

RESEARCH EXPERIENCE

MPhil Researcher - Machine Learning & Underwater Acoustic Classification — August 2025 - Present
International Centre for Neuromorphic Systems, MARCS, Western Sydney University — *Sydney, Australia*

- Achieved 98.9% accuracy and 98.8% macro-F1 on ShipsEar using compact statistical representations of STFT/CQT/MFCC features with scikit-learn RBF-SVM models, demonstrating that engineered features could outperform more computationally complex model configurations.
- Optimised STFT, CQT, Mel and MFCC representations through parameter tuning and statistical pooling, producing an approximately 10+ percentage-point improvement over earlier feature/model configurations. [Replace with exact baseline once confirmed.]
- Achieved up to 97.7% macro-F1 using CARFAC-derived auditory features, showing that compact biologically inspired BM/IHC/LI/spike representations could remain competitive with conventional spectral features and larger learning architectures.
- Reached 72.6% accuracy / 72.3% macro-F1 on DeepShip with a compact ~157K-parameter CNN, using carefully engineered acoustic representations on a substantially larger and more challenging vessel dataset.
- Benchmarked RBF-SVM, CNN and Transformer architectures across multiple acoustic representations, showing that feature design and compact models could match or exceed more computationally intensive approaches under comparable experimental settings.
- Implemented recording-level train/validation/test controls to prevent segment leakage, and deployed lightweight inference workflows to Raspberry Pi to evaluate model-size, memory and runtime constraints for edge deployment.

Research Assistant — September 2024 - Present
International Centre for Neuromorphic Systems, MARCS, Western Sydney University — *Sydney, Australia*

- Designed and conducted controlled event-camera characterisation experiments to identify hot pixels, cold pixels and abnormal event activity affecting downstream computer-vision reliability.
- Developed Arduino-controlled logarithmic illumination experiments and Python analysis pipelines to quantify event-camera responses across changing lighting conditions.
- Built asynchronous event-stream processing workflows for timestamp, spatial-coordinate and polarity data, integrating offline Faery analysis with live sensor acquisition.
- Produced repeatable sensor-analysis outputs including spatial activity maps, temporal response plots, event distributions and structured experimental records for cross-researcher review.

SELECTED MACHINE LEARNING & ENGINEERING PROJECTS

FoodVision Image Classification | PyTorch, EfficientNetB2, Vision Transformer, Hugging Face

- Achieved approximately 97% image-classification accuracy with EfficientNetB2 while reducing model size by nearly 10× compared with the Vision Transformer implementation, making high-accuracy inference substantially more practical for deployment.
- Implemented a Vision Transformer from scratch in PyTorch, including patch embeddings, positional embeddings, multi-head self-attention, Transformer encoder blocks and classification head, then benchmarked it against EfficientNetB2.
- Deployed the optimised FoodVision model to Hugging Face as an interactive application supporting user-uploaded image inference.

Event-Camera Data Compression | Winter Research Scholarship — April 2025 - November 2025

- Achieved up to 5.4× lossless compression on approximately 400 MB of event-camera data by benchmarking LZ4, ZLIB, BZ2 and LZMA.
- Compared compression ratio, processing time, throughput and decompression reliability to identify practical trade-offs between storage reduction and real-time processing.
- Evaluated algorithm suitability for Raspberry Pi and edge deployment under constrained compute, memory and communication bandwidth.

ADDITIONAL EXPERIENCE

Customer Service Coordinator — November 2021 - Present
UniLodge — *Sydney, Australia*

- Supported accommodation operations, reporting and structured records across university residences serving 1,600+ residents, coordinating with university teams, contractors, residents and staff.

EDUCATION

Western Sydney University — Sydney, Australia
Master of Philosophy — *August 2025 - Present*

- Fully funded competitive research scholarship; research focused on machine learning, underwater acoustics, neuromorphic sensing, lightweight models and embedded AI.

Western Sydney University — Sydney, Australia
Bachelor of Engineering (Honours), Electrical Engineering - GPA: 3.902 — *Completed July 2025*

PUBLICATION & ENGAGEMENT

- Co-authored a 2025 IEEE publication on drone-based sound-source localisation; publication details available through Google Scholar.
- Served as a Student Ambassador, Master of Ceremonies and Student Voice Committee representative at Western Sydney University.